

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-CT-7801-06 Sheet 1 of 1
OPL Title	Gearbox Vibration Monitoring (VSHH-7802)	Discipline	Instrument	
Equipment	CT-7801 - COOLING TOWER CELL FAN	Area / Unit	7800 - COOLING TOWER SYSTEM	
Related Interlock	SEQ-7801 (SIL 1)	P&ID; Ref	TJC-LLD-PID-7801	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **gearbox vibration monitoring (vshh-7802)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Trend the VS-7802 reading on the PdM route	Loop isolated / under permit
2	Alarm at 6 mm/s and trip at 9 mm/s (VSHH-7802)	Reference standard traceable
3	Correlate the rise with gearbox oil temperature TE-7803	As-found & as-left recorded
4	Inspect the floating drive-shaft couplings	Trip / control action verified
5	Escalate a step-change to reliability	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled pH analyzer calibration	Electrode drift	Recalibrated AI-7805 on buffer solutions, replaced electrode
Blowdown conductivity reading unreliable	Conductivity cell fouled with scale	Cleaned and recalibrated AI-7806 conductivity cell
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level

6. KEY LEARNING POINTS

- A rising vibration trend warns of gear or bearing wear before the fan trips or throws a blade.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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